

ABOUT

E&TC undergraduate specializing in bare-metal C on ARM Cortex-M, Embedded Linux on aarch64, RTL design in SystemVerilog, and QNX Neutrino RTOS. Experienced in BSP validation, IPC latency characterization, and safety-critical timing analysis on resource-constrained SoCs. Operates across the full embedded stack from RTL verification and register-level bring-up to userspace daemon engineering and live system debugging on production hardware. Proficient in building real-time systems; familiar with ASIL-D timing constraints.

EXPERIENCE

Industry Sponsored Project : IFM Electronic

Embedded Linux BSP Validation / GStreamer, Shell Scripting, Python, RTSP/RTP, Linux VFS / Programmable Logic Controller

- Validated embedded Linux BSP on CODESYS-based IFM CR1204 HMI (aarch64 Mali400 SoC, kernel 5.10.127); interfaced with PLC/HMI stack and bypassed Weston 8.0.0 IVI shell incompatibility; routed dual-stream GStreamer 1.16.3 pipeline directly to /dev/fb0, achieving stable side-by-side RTSP output without compositor.
- Resolved ARP IP conflict via tcpdump; engineered automated telemetry daemon in Shell/Python polling /proc/stat and /sys/class/net reducing peak CPU utilization to 22.6% on a VPU-less device (CPU: 0.26%, RAM: 343 MB, Pandas-validated).

PROJECTS

QNX Neutrino IPC Latency Benchmarking: Real-Time IPC Analysis (C, Python, QNX Neutrino 8.0, QEMU):

- Cross-compiled and benchmarked MsgSend/MsgReceive and POSIX shared memory + semaphore IPC mechanisms on x86-64 QEMU across 5 payload sizes (8B–16KB), 10,000 iterations each; computed p50/p99/p99.9 latency percentiles.
- Plotted latency distributions and worst-case curves in Python/matplotlib; validated real-time determinism against a simulated ASIL-D timing budget, exposing IPC overhead characteristics relevant to safety-critical scheduling.

RONAN: A Neural Network-Enhanced RISC-V Processor (SystemVerilog, Cocotb, Verilator):

- Quad-core RISC-V with hybrid (perceptron + TAGE)-based branch prediction achieving 99.89% accuracy and 0.999 IPC (vs. 0.508 baseline).
- Functionality validated via parameterized Cocotb/Verilator testbenches across 5 structured workloads.

KRONOS: Preemptive RTOS Kernel from Scratch (Bare-Metal C, ARM Assembly, IAR EWARM, CMSIS):

- Built a preemptive ARM Cortex-M4 (STM32F407VG) RTOS kernel in bare-metal C/ARM assembly with PendSV context switching, PSP/MSP separation, and SysTick priority round-robin scheduling for 4 tasks.
- Implemented MPU task isolation, MemManage fault logging (CFSR + stacked PC), DWT CYCCNT profiling, and stack high-water mark detection; 8,932-byte binary with 0 errors/warnings on IAR EWARM 9.70.4..

SKILLS

Languages: C/C++, Python, Verilog/SystemVerilog, Shell, Assembly (RISC-V, 8051)
Hardware & Platforms: STM32 (ARM Cortex-M), Jetson Nano, ESP32, Raspberry Pi, aarch64 SoCs, Logic Analyzers, PLC, TI Stellar Launchpad, Nexys 4 DDR FPGA, Vicharak Shrike Lite FPGA
Tools & OS: QNX Neutrino, Embedded Linux (User/Kernel), FreeRTOS, Git, GDB, tcpdump, Momentics IDE, STM32CubeIDE, Vivado
Protocols: UART, SPI, I²C, CAN, APB

EDUCATION

Degree	Institute/Board	CGPA/Percentage	Year
Bachelor of Technology	Vishwakarma Institute Of Information Technology	8.41	2023-2027
Senior Secondary	(MSBSHSE) Class 12 (HSC)	73.17%	2023
Secondary	(CBSE) Class 10 (SSC)	94.8%	2021

PUBLICATIONS

- A. Phadke et al., "RONAN: A Neural Network-Inspired Pipelined RISC-V Processor with Perceptron-Based Branch Prediction," Under Review, VIIT Pune, 2026.

CERTIFICATIONS

- Embedded Systems using C EDUCBA2025
- ARM Cortex (STM32) Fundamentals: Building Embedded Systems EDUCBA2025
- ARM Cortex (STM32) Deep Dive: Mastering Embedded Systems EDUCBA2025
- Introduction to Microprocessors ARM2025
- Using Git for Distributed Development The Linux Foundation2025
- Modeling and Simulation with Simulink MathWorks2025

LEADERSHIP EXPERIENCE & ACHIEVEMENTS

General Manager, IEEE VIIT Student Branch

October 2024 – Present

- Developed comprehensive curriculum for technical workshops covering Python, IoT, and sensor integration, directly impacting the learning outcomes for 100+ high school students.
- Successfully coordinated the IEEE ICBDS Conference 2024.

IEEE IoT Hackathon

- Guided a junior team to secure 2nd place by designing and implementing an automated, real-time performance monitoring system for embedded sensors, achieving complete hardware-software integration under strict time constraints.